



POLITECNICO
MILANO 1863

WellPlate: Design & Architecture

Design and Implementation of Mobile Applications exam

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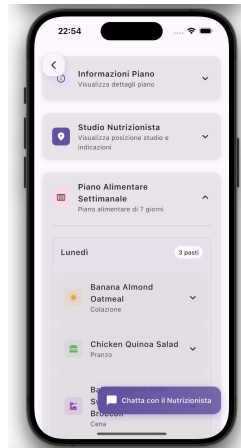
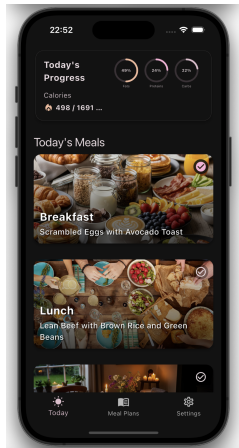
September 11, 2025

- ① Motivation & Purpose
- ② Requirements
- ③ Architecture Overview
- ④ UX & Responsiveness
- ⑤ Testing
- ⑥ Testing
- ⑦ Future Work

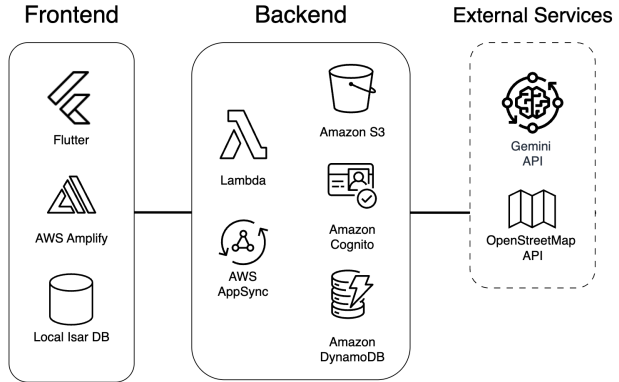
- **Problem:** Current meal-planning apps offer low personalization and limited adaptation
- **Goal:** AI-powered meal plan generation with optional professional validation
- **Target Users:** Health-conscious individuals with diverse dietary needs

- **Registration & Authentication:** users, nutritionists; Google SSO
- **Profile Setup:** demographics, allergens (14 categories), preferences
- **AI-Powered Meal Plan:** weekly generation with detailed recipes & macronutrients
- **Daily Tracking:** completion status, calories, progress
- **Pro Features:** nutritionist validation, real-time chat
- **In-App Notifications:** real-time updates for users and nutritionists

- **Usability:** intuitive UX from tablet and smartphone
- **Security:** safeguards against prompt injection in AI requests
- **Compatibility:** iOS 14+, Android 8+, iPadOS 14+
- **Accessibility:** bilingual (IT/EN), dark/light themes



-  **UI:** Flutter
-  **Data:** DynamoDB & Isar - Profiles, meal plans, chat, tracking
-  **API:** AppSync GraphQL & REST endpoints
-  **Auth:** Cognito & Google Sign-In
-  **Storage:** S3 - Nutritionist profile pictures
-  **LLM:** Google Gemini - AI meal plan generation
-  **Maps:** OpenStreetMap - Studio location services

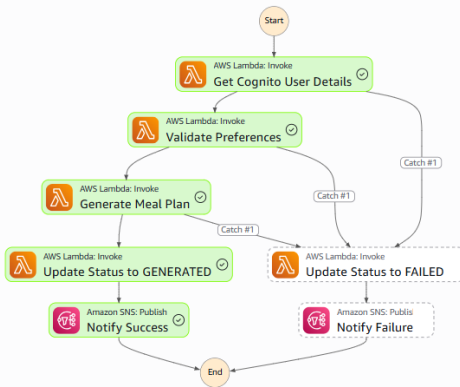


- **User Management:** UserDetails, NutritionistProfile, SubscriptionStatus
- **Meal Planning:** MealPlan → DailyPlanData → Meal → Ingredient / Macros
- **Progress Tracking:** TodaysPlan, PlanDayCompletion, MealWithStatus
- **Chat:** ChatMetadata, ChatMessage

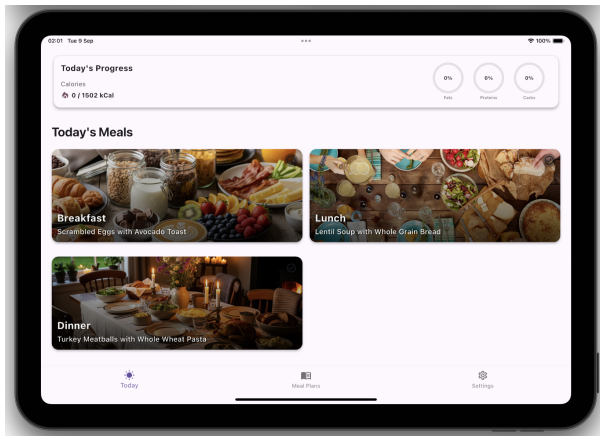
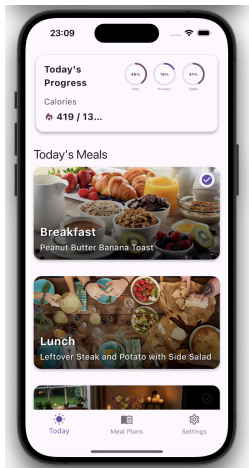
Google Gemini 2.5 Flash

- **Inputs:** profile data, allergens, preferences, constraints
- **Outputs:** weekly meal plan (recipes, macros, ingredients)
- **Structured Output:** JSON schema for consistency

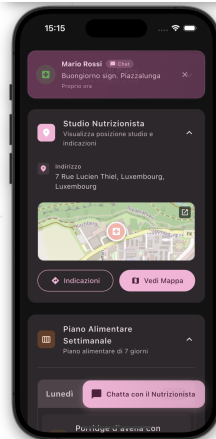
Real-time notification: user alerted once generation completes



- **User Views:** Today • My Plans • Meal Plan Viewer • Plan Generation • Chat • Settings
- **Nutritionist Views:** Validate Plans • Meal Plan Editor • Chat • Settings
- **Responsive Layouts:** Optimized for smartphones & tablets, multi-panel support on large screens
- **Robust UI States:** Graceful handling of errors, empty views, loading screens, and offline mode



- **Subscriptions:** Receive meal plan updates & chat messages in real-time
- **Persistent Connections:** App maintains live updates while active



- **Environments:** Mock vs. Development.
- **Unit tests:** models, providers, utilities, client side business rules.
- Library generated code excluded from coverage.
- **User testing:** qualitative feedback from 8 participants.

File	Lines	Line Coverage
- All Files	2753/3089	89.1%
- lib	2753/3089	89.1%
+ models	550/562	97.9%
+ providers	838/926	90.5%
+ services	1251/1474	84.9%
+ Utils	114/127	89.8%

Figure: Unit Testing coverage

- **Enhanced AI:** deeper personalization, CV-based meal logging, NLP assistants
- **Wearable Integration:** HealthKit, Google Fit, Fitbit for closed-loop plans
- **Community & Analytics:** features for users and professional insights for nutritionists
- **Performance Improvements:** caching, offline support, and optimization

Thank you!

Live Demo